EX No:02

Date:

Exploring College Dataset to Find Insights

2. This exercise relates to the College data set, which can be found in the file College.csv on the book website. It contains a number of variables for 777 different universities and colleges in the US. The variables are

LOAD LIBRARY:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from ISLP import load_data
from matplotlib.pyplot import subplot
```

(a) Use the pd.read_csv() function to read the data into Python. Call the loaded data college. Make sure that you have the directory set to the correct location for the data.

LOAD DATA:

```
college=pd.read_csv('College.csv')
```

(b) Look at the data used in the notebook by creating and running a new cell with just the code college in it. You should notice that the first column is just the name of each university in a column named something like Unnamed: 0. We don't really want pandas to treat this as data. However, it may be handy to have these names for later. Try the following commands and similarly look at the resulting data frames.

This has used the first column in the file as an index for the data frame. This means that pandas has given each row a name corresponding to the appropriate university. Now you should see that the first data column is Private. Note that the names of the colleges appear on the left of the table. We also introduced a new python object above: a dictionary, which is specified by dictionary (key, value) pairs. Keep your modified version of the data with the following:

CODE:

```
college = college3
college2 = pd.read_csv('College.csv', index_col=0)
college3 = college.rename({'Unnamed: 0': 'College'},axis=1)
college=college3
college.head()
```

OUTPUT:

	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.Ratio
0	Yes	1660	1232	721	23	52	2885	537	7440	3300	450	2200	70	78	18.1
1	Yes	2186	1924	512	16	29	2683	1227	12280	6450	750	1500	29	30	12.2
2	Yes	1428	1097	336	22	50	1036	99	11250	3750	400	1165	53	66	12.9
3	Yes	417	349	137	60	89	510	63	12960	5450	450	875	92	97	7.7
4	Yes	193	146	55	16	44	249	869	7560	4120	800	1500	76	72	11.9

(c) Use the describe() method of to produce a numerical summary of the variables in the data set.

CODE:

```
college.describe()
```

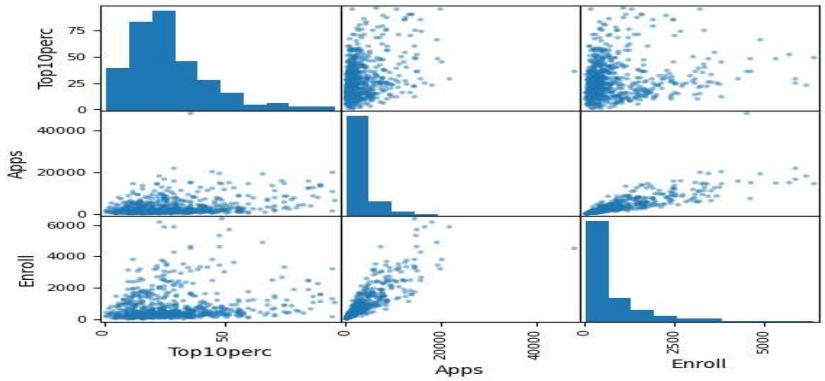
OUTPUT:

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal
count	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000
mean	3001.638353	2018.804376	779.972973	27.558559	55.796654	3699.907336	855.298584	10440.669241	4357.526384	549.380952	1340.642214
std	3870.201484	2451.113971	929.176190	17.640364	19.804778	4850.420531	1522.431887	4023.016484	1096.696416	165.105360	677.071454
min	81.000000	72.000000	35.000000	1.000000	9.000000	139.000000	1.000000	2340.000000	1780.000000	96.000000	250.000000
25%	776.000000	604.000000	242.000000	15.000000	41.000000	992.000000	95.000000	7320.000000	3597.000000	470.000000	850.000000
50%	1558.000000	1110.000000	434.000000	23.000000	54.000000	1707.000000	353.000000	9990.000000	4200.000000	500.000000	1200.000000
75%	3624.000000	2424.000000	902.000000	35.000000	69.000000	4005.000000	967.000000	12925.000000	5050.000000	600.000000	1700.000000
max	48094.000000	26330.000000	6392.000000	96.000000	100.000000	31643.000000	21836.000000	21700.000000	8124.000000	2340.000000	6800.000000

(d) Use the pd.plotting.scatter_matrix() function to produce a scatterplot matrix of the first columns [Top10perc, Apps, Enroll]. Recall that you

CODE:

```
pd.plotting.scatter_matrix(college[['Top10perc','Apps','Enroll']]);
```

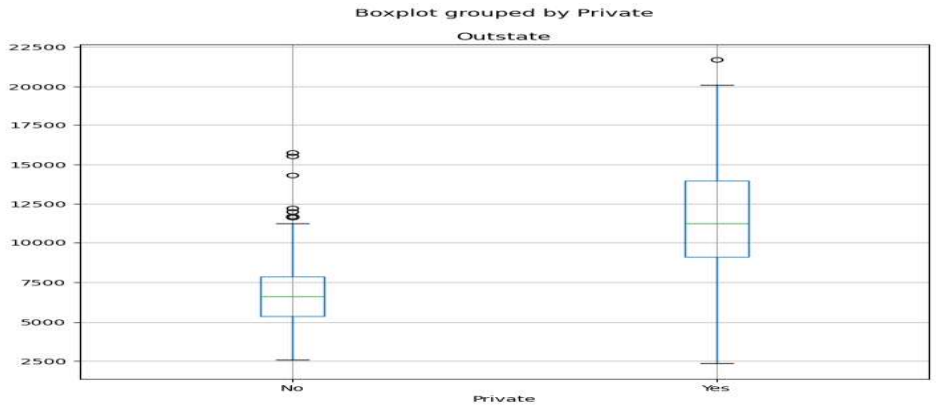


(e) Use the boxplot() method of college to produce side-by-side boxplots of Outstate versus Private.

CODE:

```
fig, ax = plt.subplots(figsize=(8, 8))
college.boxplot('Outstate', by='Private', ax=ax);
```

OUTPUT:



(f) Create a new qualitative variable, called Elite,bybinning the Top10perc variable into two groups based on whether or not the proportion of students coming from the top 10% of their high school classes exceeds 50%.

college['Elite'] = pd.cut(college['Top10perc'], [0,0.5,1], labels=['No', 'Yes']) Use the value_counts() method of college['Elite'] to see how many elite universities there are. Finally, use the boxplot() method again to produce side-by-side boxplots of Outstate versus Elite.

CODE:

```
college['Elite'] = pd.cut(college['Top10perc'],
      [0, 50, 100],
      labels=['No', 'Yes'])
college['Elite'].value_counts()
```

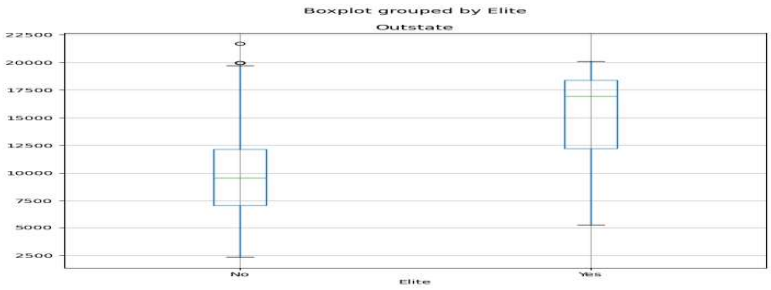
OUTPUT:

```
Elite
No      699
Yes      78
Name: count, dtype: int64
```

CODE:

```
fig, ax = plt.subplots(figsize=(8, 8))
college.boxplot('Outstate', by='Elite', ax=ax);
```

OUTPUT:

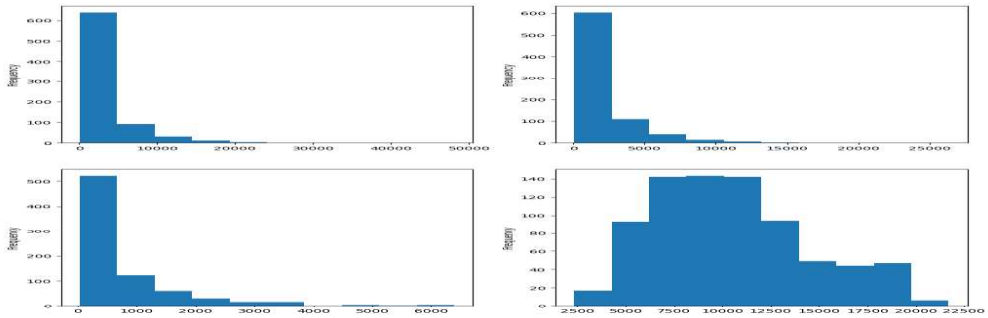


(g) Use the plot.hist() method of college to produce some histograms with differing numbers of bins for a few of the quantitative variables. The command plt.subplots(2, 2) may be useful: it will divide the plot window into four regions so that four plots can be made simultaneously. By changing the arguments you can divide the screen up in other combinations.

CODE:

```
fig, axes =plt.subplots(nrows=2,ncols= 2,figsize=(12,12))
college['Apps'].plot.hist(ax=axes[0,0])
college['Accept'].plot.hist(ax=axes[0,1])
college['Enroll'].plot.hist(ax=axes[1,0])
college['Outstate'].plot.hist(ax=axes[1,1]);
```

OUTPUT:



(h) Continue exploring the data, and provide a brief summary of what you discover.

CODE:

```
college.info()
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 777 entries, 0 to 776
Data columns (total 19 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Private      777 non-null    object
 1   Apps         777 non-null    int64
 2   Accept       777 non-null    int64
 3   Enroll       777 non-null    int64
 4   Top10perc    777 non-null    int64
 5   Top25perc    777 non-null    int64
 6   F.Undergrad  777 non-null    int64
 7   P.Undergrad  777 non-null    int64
 8   Outstate     777 non-null    int64
 9   Room.Board   777 non-null    int64
 10  Books        777 non-null    int64
 11  Personal     777 non-null    int64
 12  PhD          777 non-null    int64
 13  Terminal     777 non-null    int64
 14  S.F.Ratio    777 non-null    float64
 15  perc.alumni  777 non-null    int64
 16  Expend       777 non-null    int64
 17  Grad.Rate    777 non-null    int64
 18  Elite        777 non-null    category
dtypes: category(1), float64(1), int64(16), object(1)
memory usage: 110.3+ KB
```

CODE:

```
college['Private']=college['Private'].astype('category')

# Correlations among key variables
college[['Outstate', 'Top10perc', 'Grad.Rate', 'PhD', 'S.F.Ratio', 'Expend']].corr()
```

OUTPUT:

	Outstate	Top10perc	Grad.Rate	PhD	S.F.Ratio	Expend
Outstate	1.000000	0.562331	0.571290	0.382982	-0.554821	0.672779
Top10perc	0.562331	1.000000	0.494989	0.531828	-0.384875	0.660913
Grad.Rate	0.571290	0.494989	1.000000	0.305038	-0.306710	0.390343
PhD	0.382982	0.531828	0.305038	1.000000	-0.130530	0.432762
S.F.Ratio	-0.554821	-0.384875	-0.306710	-0.130530	1.000000	-0.583832
Expend	0.672779	0.660913	0.390343	0.432762	-0.583832	1.000000

CODE:

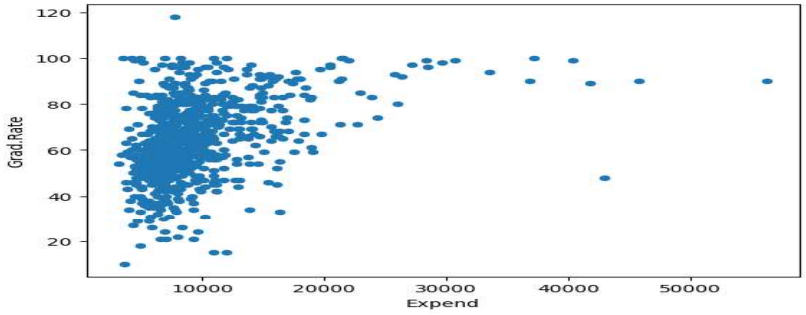
```
college.groupby(['Private', 'Elite'])['Grad.Rate'].mean()
```

OUTPUT:

```
Private  Elite
No       No       54.643216
         Yes      77.461538
Yes      No       66.974000
         Yes      84.569231
Name: Grad.Rate, dtype: float64
```

CODE:

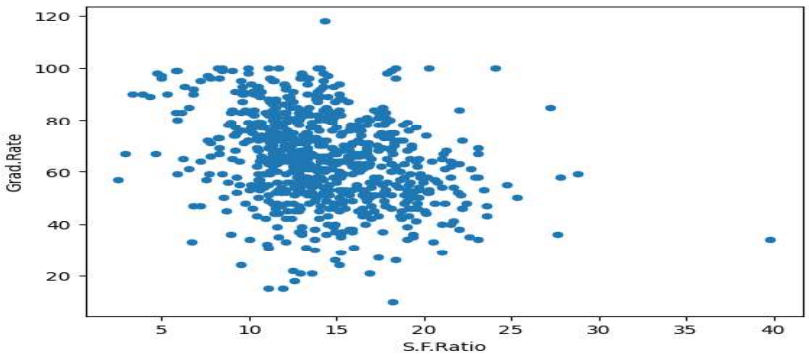
```
# Spending vs. graduation
college.plot.scatter(x='Expend', y='Grad.Rate')
plt.show()
```



CODE:

```
college.plot.scatter(x='S.F.Ratio', y='Grad.Rate')
plt.show()
```

OUTPUT:



INTERPRETATION:

- The College data holds 777 colleges. Reading the file with the first column as the index (and renaming it College) keeps the college names available without treating them as data.
- (c) The numerical summary shows strong right-skew in the size variables: Apps has mean 3,002 against median 1,558 (max 48,094), Enroll mean 780 against median 434, and F.Undergrad mean 3,700 against median 1,707. A few very large universities pull the means well above the medians. Outstate is far more symmetric (mean 10,441, median 9,990).
- (d) The scatterplot matrix shows Apps and Enroll rising together, i.e. larger institutions receive more applications and enrol more students.
- (e) Private colleges charge clearly higher out-of-state tuition than public colleges: the Outstate boxplot for Private = Yes sits well above the one for Private = No.
- (f) Only 78 of the 777 colleges (about 10%) are Elite (more than 50% of new students from the top 10% of their class); 699 are not. Elite colleges also have noticeably higher Outstate tuition than non-elite ones.
- (g) The histograms of Apps, Accept and Enroll are strongly right-skewed (many small colleges, very few huge ones), whereas Outstate is much closer to bell-shaped.
- (h) Graduation rate is positively correlated with Outstate (0.57), Top10perc (0.49), Expend (0.39) and PhD (0.31), and negatively with S.F.Ratio (-0.31). Expend is also strongly related to Outstate (0.67) and Top10perc (0.66), and S.F.Ratio is negatively related to Expend (-0.58) and Outstate (-0.55), so better-funded colleges have smaller classes and higher graduation rates.
- Mean graduation rate by group: public non-elite 54.6%, public elite 77.5%, private non-elite 67.0%, private elite 84.6%. Elite status adds roughly 18 to 23 percentage points and being private adds roughly 7 to 12 points. The scatterplots agree: graduation rate rises with spending per student and falls as the student/faculty ratio rises.